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PERSONAMEM-V2: Towards Personalized Intelligence via Learning Implicit User Personas and Agentic Memory

Bowen Jiang[1], Yuan Yuan[1], Maohao Shen[2], Zhuoqun Hao[1], Zhangchen Xu[3], Zichen Chen[4], Ziyi Liu[6], Anvesh R. Vijjini[7], Jiashu He[1], Hanchao Yu[5], Radha Poovendran[3], Gregory Wornell[2], Lyle Ungar[1], Dan Roth[1], Sihao Chen[8]*, Camillo J. Taylor[1]*

University of Pennsylvania[1], Massachusetts Institute of Technology[2], University of Washington[3], University of California Santa Barbara[4], Meta[5], University of Southern California[6], University of North Carolina at Chapel Hill[7], Microsoft Corporation[8]

bwjiang@seas.upenn.edu, sihaochen@microsoft.com, cjtaylor@seas.upenn.edu

🤗 *huggingface.co/datasets/bowen-upenn/PersonaMem-v2*

Abstract

Personalization is one of the next milestones in advancing AI capability and alignment. We introduce PERSONAMEM-V2, the state-of-the-art dataset for LLM personalization that simulates 1,000 realistic user–chatbot interactions on 300+ scenarios, 20,000+ user preferences, and 128k-token context windows, where most user preferences are implicitly revealed to reflect real-world interactions. Using this data, we investigate how reinforcement fine-tuning enables a model to improve its long-context reasoning capabilities for user understanding and personalization. We also develop a framework for training an agentic memory system, which maintains a single, human-readable memory that grows with each user over time.

In our experiments, frontier LLMs still struggle with implicit personalization, achieving only 37–48% accuracy. While they support long context windows, reasoning remains the bottleneck for implicit personalization tasks. Using reinforcement fine-tuning, we successfully train Qwen3-4B to outperforms GPT-5, reaching 53% accuracy in implicit personalization. Moreover, our agentic memory framework achieves state-of-the-art 55% accuracy while using 16× fewer input tokens, relying on a 2k-token memory instead of full 32k conversation histories. These results underscore the impact of our dataset and demonstrate agentic memory as a scalable path toward real-world personalized intelligence.

1 Introduction

Personalization is becoming one of the next milestones towards artificial superintelligence (OpenAI, 2025;b; Meta, 2025; Chowdhury, 2025; Evolution AI Hub, 2025). As the user base of artificial intelligence grows dramatically in both size and diversity, large language models (LLMs) increasingly face the challenge of serving users with distinct personas and backgrounds. In many real-world applications, such as education, healthcare, and empathetic emotional support (Jiang et al., 2025; Ghimire et al., 2024; Ivanovic et al., 2022; Gómez-González et al., 2020; Baillifard et al., 2025; Schaaff et al., 2023), there is no single "correct" answer. Instead, success depends on delivering personalized responses that align with individual users' intentions, contexts, preferences, and emotional states. **Personalization offers a path toward pluralistic alignment**, shifting the reasoning goals towards factual correctness to meaningful resonance across diverse users.

*Equal advising.

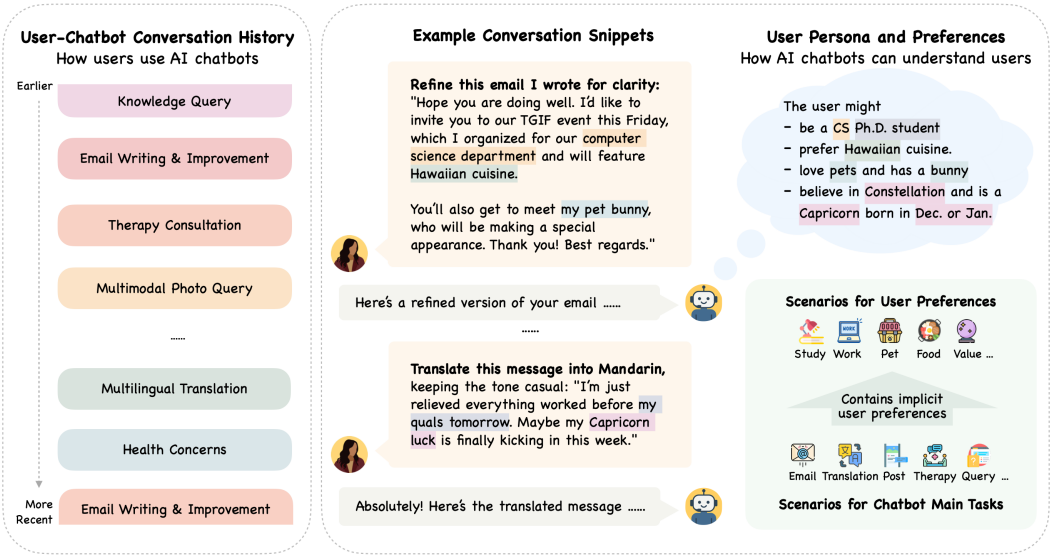

Figure 1: Overview of PERSONAMEM-V2 benchmark. On the left, we mimic how people talk to chatbots across many topics over time, forming long and varied interaction histories. In the middle, we zoom in on a few conversation snippets. Even simple task scenarios like refining an email or translating a message naturally reveal small details about someone's life in many other scenarios, like what they study, what foods they like, whether they have pets, or what they're planning for tomorrow. Taken together, these subtle signals help models build thorough yet succinct user profiles for personalization.

A key enabler of this vision is the growing history of user–chatbot interactions, which encode rich signals about user preferences. However, personalization in LLMs remains a challenge. **Most users do not explicitly state their preferences to chatbots.** According to OpenAI's recent report (OpenAI, 2025), most users still treat LLMs as tools, meaning most of their preferences are revealed only implicitly through everyday interactions. For example, someone might ask a chatbot to help polish the writing of an email, but the email itself could reveal their dining habits, as shown in Figure 1. In this case, the chatbot needs to pick up on those preferences while carrying out the main task the user asked for. Furthermore, many users engage in exploratory or hypothetical prompts to test the model's capabilities, which can obscure the true preferences of their own. As a result, real-world user-chatbot conversation histories tend to be long and noisy, requiring models to infer implicit user personas and preferences from scattered, indirect evidence over time.

This raises a central question: **how well can LLMs understand the users, especially their implicit personas and preferences from long conversation histories, and therefore provide personalized responses?**

To this end, we present the state-of-the-art LLM personalization dataset, PERSONAMEM-V2: IMPLICIT PERSONAS, that captures the complexity of real-world user–chatbot interactions. The dataset spans over 1,000 user personas, covering comprehensive demographic attributes, mental health and medical backgrounds, as well as stereotypical, anti-stereotypical, and neutral user preferences. In addition, **it features realistic, dynamic, multimodal, multilingual, and multi-session user–chatbot conversation histories that implicitly convey user preferences**, encompassing over 20,000+ user preferences over 300+ everyday conversation topics, and up to 128,000 tokens per context. To support both evaluation and model training, the dataset includes 5,000 high-quality Q&A pairs curated for benchmarking and an additional 20,000 Q&A pairs for training and validation, each generated through a scalable, multi-step validation pipeline that enforces strong quality standards.

To understand how well today's most capable models personalize to real users, we started by testing frontier LLMs from OpenAI (OpenAI, 2025a; 2024b;a) on these implicit signals

that show up in everyday conversations. These evaluations quickly revealed a gap: even the strongest models among them struggle to interpret and track user preferences over long interaction histories. **This gap motivated us to explore the value of our data in driving the next wave of personalized intelligence.** Using reinforcement fine-tuning (RFT) with GRPO (Guo et al., 2025), we find that RFT is remarkably effective for personalization, even though personalization is inherently subjective and open-ended. Our dataset enables us to train a 4B-parameter reasoning model capable of long-context reasoning for personalization.

Another key contribution of our work is an agentic memory framework that learns to build and update a persistent, human-readable memory that evolve with each user. Instead of relying on full conversational transcripts, the model is trained via reinforcement fine-tuning to distill long interaction histories into a compact 2k-token memory that captures the user's evolving personas and preferences. This memory becomes the model's sole personalization context, allowing it to reason over implicit signals and maintain strong user understanding even across long, multi-session conversations. Despite using 16× fewer input tokens, the agentic memory model delivers state-of-the-art performance on implicit personalization, surpassing frontier LLMs on both multiple-choice and open-ended evaluations. This demonstrates a scalable path toward AI systems that remember what matters for each user, infer subtle user preferences, and adapt over time to give personalized responses, all with the efficiency required for real-world deployment.

To summarize our contributions:

- We curate the state-of-the-art dataset for LLM personalization, featuring realistic user–chatbot interactions that reveal implicit user preferences.
- We benchmark the implicit personalization capabilities of frontier LLMs.
- We demonstrate the efficacy of reinforcement fine-tuning for personalization.
- We propose an agentic memory framework that achieves state-of-the-art performance and efficiency, outperforming GPT-5 in personalization.

2 Overview of PersonaMem-v2: Implicit Personas

2.1 Comprehensive User Personas

PersonaMem-v2 introduces **1,000 richly detailed personas** that capture nearly the full spectrum of demographic diversity across global regions, cultures, races, genders, and sexual orientations, reflecting the pluralism that real-world AI systems must serve. Each persona draws from a random description in PersonaHub (Ge et al., 2024) and expands it into an unfixed set of attributes that cover, but are not limited to, professional and educational backgrounds, personal characteristics, relationships, values and beliefs, technological familiarity, and conversational styles. Moreover, as people increasingly rely on chatbots for personal use and care, PersonaMem-v2 specifically includes mental and physical health backgrounds where personalization carries high stakes.

A central question for LLM personalization is what a chatbot should remember about the user. The chatbot should remember what the user has shared about themselves, whether conveyed implicitly or explicitly, rather than relying on static assumptions or stereotypes derived solely from demographic attributes. To explore this boundary, PersonaMem-v2 provides **stereotypical, anti-stereotypical, and neutral user preferences** for each persona, as well as health- and therapy-related ones, totaling around 20,000+, enabling systematic evaluation of how models ground personalization in conversational evidence.

Chatbots often misinterpret user behavior. This can occur when people test the model with hypothetical examples to explore its capabilities, or with messages written by a third person, and the model fails to distinguish them from the user's own persona. PersonaMem-v2 addresses this challenge by including such ambiguous cases. Besides, we also account for the **dynamic nature of user preferences and how they evolve over time** across multiple sessions. For instance, a user who initially expresses interest in vegetarian recipes might later ask for high-protein meal suggestions after starting a new fitness routine.